Lab 3: Component Modelling and Architectural Pattern Selection

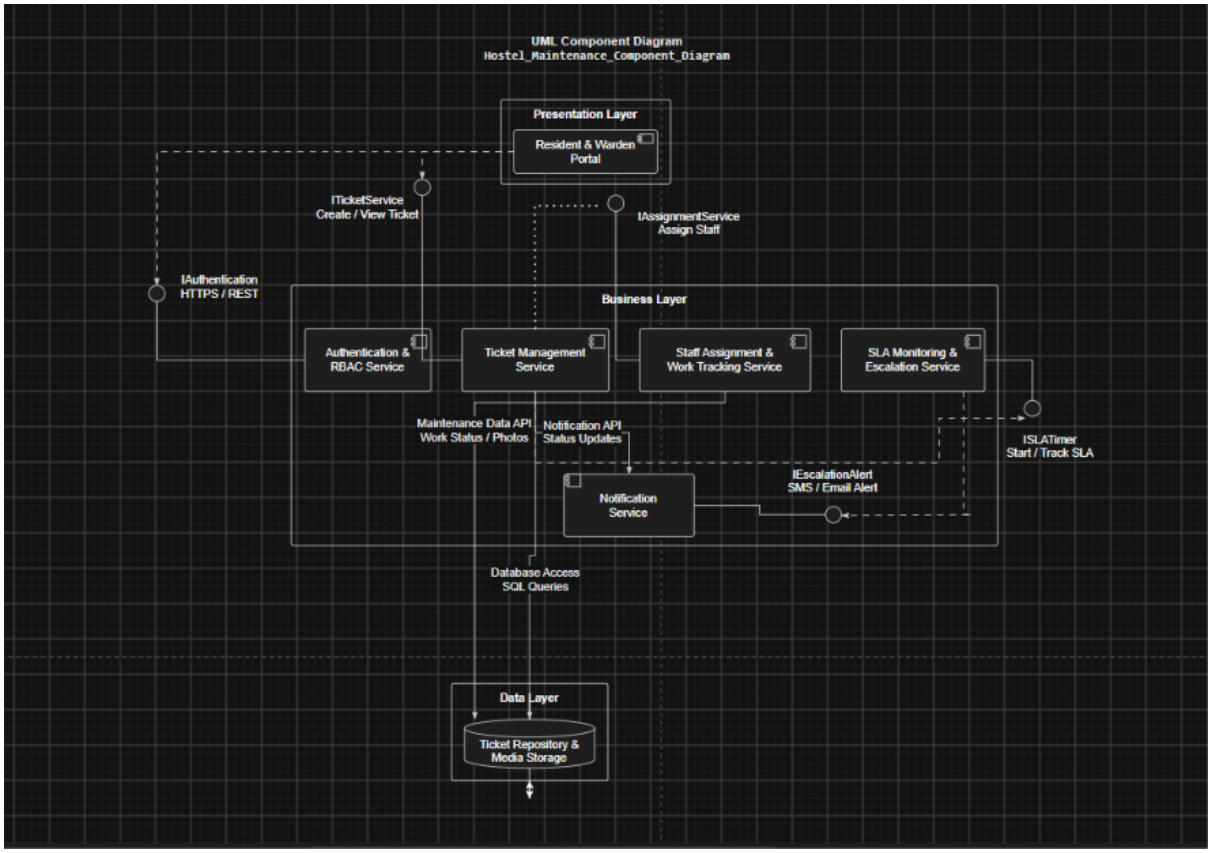
Hostel Maintenance & Issue Ticketing System

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Section: D

- **Component Diagram**



- **Architecture Selection and Justification**

We chose Layered Architecture for the Hostel Maintenance & Issue Ticketing System.

The first reason is that layered architecture provides a clear separation of concerns. The Presentation Layer contains the Resident and Maintenance Warden portal. The Business Layer manages ticket creation, staff assignment, status updates, SLA monitoring, escalation, and resident closure verification. The Data Layer stores ticket information and encrypted issue photos. This separation makes the system easier to maintain and modify.

The second reason is that the system is suitable for a centralized hostel-management environment. Residents can raise and track maintenance tickets, while the Maintenance Warden can review tickets, assign staff, and monitor deadlines through one controlled system.

A security advantage is centralized Authentication and Role-Based Access Control. Residents can access only tickets belonging to their own room, while Wardens and maintenance staff receive access according to their roles. Communication can use TLS, and photo uploads and credentials can be stored securely.

A performance benefit is that SLA Monitoring and Escalation can run in the background. This allows residents to submit tickets and view their ticket status without waiting for SLA checks or notification processing to finish.